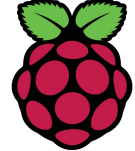


Embedded Systems

Course introduction

winter 2026

Basic information

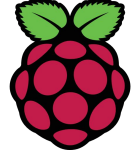


- classes - 32h
- meetings - 8 x 4h
- laboratory - B109P (build. B)
- recommended IDE - thonny.org

Th



Hardware



Raspberry Pi
Pico



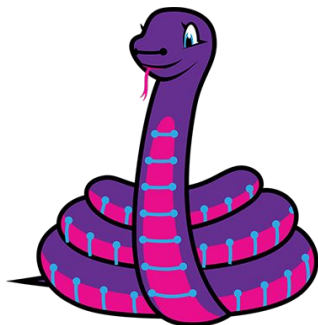
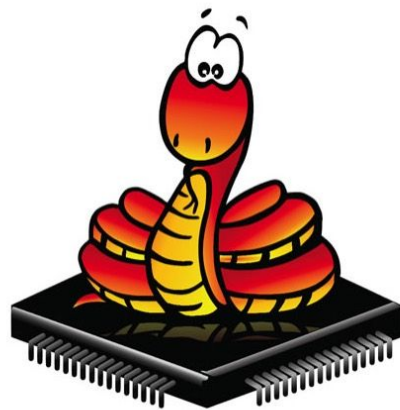
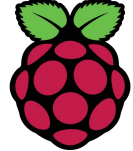
Raspberry Pi
Pico 2 W

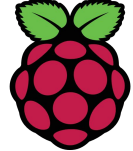


Raspberry Pi 4B

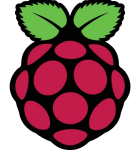


Programming language





- **class #1** - Raspberry Pi Pico - General Purpose
Inputs/Outputs, Multi Thread Coding
- **class #2** - Raspberry Pi Pico - Analog inputs, UART,
Asynchronous I/O Scheduler, Files
- **class #3** - Raspberry Pi Pico - Interrupts

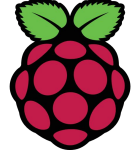


- **class #4** - Raspberry Pi Pico W - Networking
- **class #5** - Raspberry Pi Pico W - Communication

Interfaces

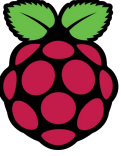
- **class #6** - Raspberry Pi Pico W - IOT, MQTT

Timetable



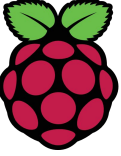
- **class #7** - Final project
- **class #8** - Final project

Grades - option 1 - solve tasks (individual)



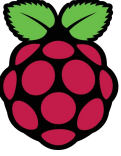
- 3.0 - present a solution for one unsolved task from every manual - class 1 to 4;
- 3.5 - present a solution for one unsolved task from every manual - class 1 to 5;
- 4.0 - present a solution for one unsolved task from every manual - class 1 to 6.

Grades - option 1 - mini project (group of 2)



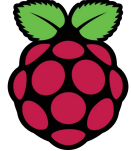
- 5.0 - final project (classes 7 and 8).

Grades - option 2 - project path



- individual project;
- deadline #1 - 30.11.2026 - design;
- deadline #2 - 28.01.2027 - project presentation;
- grade depends on quality of a presented solution and answers for questions about the solution;

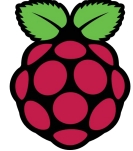
Grades - summary



- deadline - 28.01.2026
- 2 tasks per meeting
- presentation is mandatory
- sending the solution is mandatory

Do not send files with the .py extension

Classes organization



- **1st hour** - own work - solutions presentation
- **2nd hour** - new topic introduction
- **3rd hour** - own work - time to solve tasks
- **4th hour** - own work - time to solve tasks